# Diego R. Caputo

PhD Student, Industrial & Systems Engineering | Lean Manufacturing | Six Sigma | AI in Industrial Engineering

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## Summary

PhD student in Industrial and Systems Engineering at Auburn University (expected 2027) with 10 years of experience across engineering, data analytics, and technical instruction. Skilled at breaking complex technical processes into clear, structured, verifiable content, and at reviewing work against defined quality criteria. Builds data tools and software in Python and R using AI-assisted development workflows. Experienced writing and teaching technical material for students and industry teams in English and Spanish.

## Skills

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| **AI and data** | AI-assisted software development, data quality review, data analysis, data modeling, ETL pipelines, dashboard development, technical writing |
| **Programming and tools** | Python, R, Microsoft Excel, Power BI, Minitab, 3DVista |
| **Engineering and process** | Lean manufacturing (5S, Kanban, SMED, Kaizen, A3), Six Sigma, DMAIC, OEE, root cause analysis, process mapping, value stream mapping, continuous improvement |
| **Training and instruction** | Curriculum and training module design, instructional content development, student assessment and project supervision, technical communication |
| **Languages** | English (fluent), Spanish (fluent) |

## Education

**PhD, Industrial and Systems Engineering** | Auburn University — Expected 2027

**MS, Industrial and Systems Engineering** | Auburn University | GPA 3.94 — Expected 2026

**Master of Engineering Management, Systems Option** | Auburn University | GPA 4.0 — Dec 2023

**BS, Industrial Engineering** | Pontificia Universidad Católica Madre y Maestra | GPA 3.5 — Sep 2017

## Experience

**Graduate Research Assistant** | Auburn University, Auburn, AL — Aug 2022 - Present

- Lead platform developer, Tiger Motors Lean Education Center (tigermotorslean.com): built the site, 360-degree lab tours and learning modules for a $400K NSF-funded project (DUE-2417082)
- Evaluation with 148 students raised 5S and value stream mapping quiz scores from 50-56% to 80-86%; teams from 20+ organizations, including Honda, GE Aerospace and Brose, trained with it
- Conduct systematic literature reviews, synthesizing published research into structured findings for dissertation and journal manuscripts
- Map and assess process manufacturing operations, including sawmills, to identify improvement opportunities for the Auburn University Rural Partnership Institute
- Build websites, data analytics utilities, and educational software tools using AI-assisted development workflows

**Graduate Teaching Assistant, Lean Systems (INSY 5800/6800)** | Auburn University, Auburn, AL — Aug 2022 - Present

- Run every lab and give lectures each semester for 60 to 80 students, plus about 20 students in summer project cohorts
- Run structured simulation exercises covering 5S, Kanban, just-in-time, SMED, Heijunka, A3 problem solving, and Kaizen
- Advise student Kaizen projects that raised OEE to target; evaluate student work against defined technical criteria

**Program Director, International Lean Six Sigma Green Belt** | Auburn University, USA, Germany, Spain — 2025 - Present

- Led program logistics, industry partnerships, course instruction, and supervision of student DMAIC improvement projects at German companies
- Coordinated international student cohorts in Germany and Pamplona, Spain (2025 and 2026), including a Six Sigma project on a BSH Bosch vacuum production line
- Designed a three-week Lean Six Sigma module built around a full DMAIC project; Germany 2026 graduates earned the International Lean Six Sigma Green Belt issued by Auburn University and THWS

**AI Trainer Specialist** | Outlier, Remote — Dec 2024 - May 2025

- Trained and evaluated AI models by writing and reviewing prompts and responses against defined quality guidelines
- Applied engineering and technical domain knowledge to check the accuracy of model outputs and give structured feedback
- Met project quality standards and deadlines in a self-directed, remote role

**Data Specialist** | Bank Superintendence, Santo Domingo, Dominican Republic — Mar 2021 - Dec 2021

- Designed executive dashboards to support decision making by senior leadership
- Built reusable ETL pipelines that cut data preparation time by about one week per reporting cycle

**Engineer III, Processes** | Claro Dominicana, Santo Domingo, Dominican Republic — Jun 2018 - Mar 2021

- Analyzed and automated service provisioning processes for operating system services
- Assessed system requirements and feasibility to improve service activation efficiency
- Promoted from Senior Process/Data Analyst (Sep 2017 - Jun 2018)

**Production Supervisor** | IE Industrial Electric S.A.S., Santo Domingo, Dominican Republic — Jun 2016 - Aug 2016

- Supervised about 50 plant operators, meeting production, safety, and quality requirements
- Coordinated daily production schedules under constrained resources

## + Research and Publications

### Manuscripts in progress

Caputo, D., Tran, C., Andrews, E., Ledford, A., & Harris, G. *Building Blocks for a Continuous Improvement-Driven Operational Excellence Program: A Synthesis of Systematic Literature and Industry Expert Roundtable Insights.* Manuscript prepared for submission to The TQM Journal.

Caputo, D., et al. *Using Lean Thinking to Evaluate How Lean Manufacturing Is Taught: A Systematic Review and Waste-Based Taxonomy of Training Delivery Methods.* Work-in-progress paper, ASEE Annual Conference & Exposition 2027.

Caputo, D., Gil, G., Devall, T., & Valenzuela, J. *Extending Experiential Lean Manufacturing Education Beyond the Laboratory: An Immersive Digital Platform for the Electronics Manufacturing Workforce.* Abstract submitted, SMTA Pan Pacific Strategic Electronics Symposium 2027.

### Technical reports

Tran, C., Ledford, A., Andrews, E., Caputo, D., & Harris, G. (2024). *Building an Operational Excellence Program* (ICAMS Technical Report 24-01). Auburn University.

### Presentations

Caputo, D., Tran, C., Andrews, E., Ledford, A., & Harris, G. (2025). *Strategies for Building an Operational Excellence Program.* Poster, Huntsville Research Symposium, Auburn University Research and Innovation Campus, Huntsville, AL.

Caputo, D., Tarman, W., Ballard, V., & Devall, T. (2022). *Tiger Motors Lean Education Center: Student Built, Experimental Lean Learning.* Poster, IISE Lean Six Sigma & Data Science Conference, Atlanta, GA.

## + Certifications

- International Lean Six Sigma Green Belt, Auburn University and THWS (Germany), 2026
- Lean Six Sigma Green Belt, Auburn University, 2023
- Graduate Certificate, Modeling and Data Analytics, Auburn University, 2025
- Lean Six Sigma Black Belt, Villanova University (starts 2027)

## + Honors and Leadership

- Winner, INCOSE Systems Showdown, 2026
- JT Black Lean Engineering Student of the Year, 2022
- IPC Scholarship, 2023, 2024, 2025
- President, SME Auburn University Student Chapter, 2025-2026
- Vice President, SHPE Auburn University Chapter, 2023-2024